Solution Architecture

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below shows OptiCrop's high-level architecture across the Presentation, Application, and Data layers, mapping the specific technologies and components used to deliver real-time, ML-driven agricultural recommendations to farmers.

Presentation / Client Layer
Web Dashboard (React.js) | *Farmer Mobile App (Flutter)*

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API Gateway / Load Balancer
AWS API Gateway | *Nginx Reverse Proxy*

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Auth Service	**Core Logic Service**	**External API Integrations**
JWT & OAuth2	*ML Models (Crop Recommendation, Yield Prediction), Django REST Framework*	*OpenWeatherMap, Twilio SMS, Google Maps*

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Data / Storage Layer
PostgreSQL + PostGIS | *MongoDB* | *AWS S3*

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the farmer-facing mobile experience and the agronomist/admin web dashboard for viewing recommendations, farm data, and reports.	React.js, Flutter, Tailwind CSS
API Gateway	Routes incoming requests to the correct backend service, handles authentication checks, rate limiting, and load balancing across service instances.	AWS API Gateway, Nginx

Microservices	Handles user authentication, runs the core ML models for crop recommendation and yield prediction, and integrates with third-party weather, SMS, and mapping APIs.	JWT/OAuth2, Django REST Framework, scikit-learn, TensorFlow, OpenWeatherMap API, Twilio API, Google Maps API
Database	Stores structured farmer, farm, and crop records, geospatial field boundaries, sensor/weather time-series data, and satellite imagery.	PostgreSQL with PostGIS, MongoDB, AWS S3